

University Portal

Project Feature List & Overview

Feature 01: Smart Course Recommendation System (Unique)

Description:

The Smart Course Recommendation System automatically suggests the most appropriate next courses for a student by querying their academic history stored in the database.

Example Output:

Completed Course	Recommended Course	Reason
CSE110	CSE111	Prerequisite satisfied
CSE110	CSE230	Prerequisite satisfied

Feature 02: Student Performance Analytics Dashboard (Unique)

Description:

This feature gives faculty members access to academic performance data for any course they teach. Rather than viewing individual grades in isolation, faculty can see class wide statistics pulled directly.

Example Output:

Course	Metric	Value
CSE110	Class Average	B+ (3.3)
CSE110	Highest Grade	A (4.00)
CSE110	Lowest Grade	D (1.00)

Feature 03: Department Statistics Dashboard (Unique)

Description:

The Department Statistics Dashboard allows students and administrators to view a real-time count of how many students are enrolled in each academic department.

Example Output:

Department	Total Students
Computer Science & Engineering (CSE)	200
Business Administration (BBA)	150
English (ENG)	95

Feature 04: Real-Time Lab Availability Checker (Unique)

Description:

This feature enables students to check in real time which computer labs on campus are currently free and which are occupied.

Example Output:

Lab Room	Current Status
Lab 9A-02L- 9:00 AM	Free
Lab 9A-03L-9:00 AM	Occupied
Lab 9A-04L-9:00 AM	Free

Feature 05: Campus Facility Complaint System (Unique)

Description:

The Campus Facility Complaint System provides a structured way for students to report infrastructure or equipment problems in classrooms, labs, and common areas. Students can report issues like: Broken AC, Projector not working, WIFI problem, Lab computer damaged etc.

Feature 06: Student Notes Resource Library (Unique)

Description:

The Student Notes Resource Library is a peer driven academic content sharing platform. Students can upload handwritten notes, scanned PDFs, and other study materials. Future students enrolled in the same course can browse and download these materials. Every download is logged to track, which feed directly into the Reward Points System (Feature 07).

Example Output:

Course	Note Title	Uploader	Downloads	Rating
CSE221	Handwritten Algorithm Notes	Faiyaz	73	4.7
MAT120	Calculus & Integration Notes	Fabiha	51	4.5

Feature 07: Reward Points System for Contributors (Unique)

Description:

The Reward Points System incentivizes students to share quality academic content by awarding database-tracked points for their contributions to the Notes Resource Library. Top contributors can get rewards.

Points Award Logic

Action / Trigger Condition	Points Awarded
Student uploads a note to the library	+10 points
Note receives average rating ≥ 4.5 stars	+5 points
Note download count exceeds 50	+20 points

Example Leaderboard:

Student Name	Total Points
Shuvo	120
Fabiha	90
Faiyaz	75

Feature 08: Credit Limit Checker (Basic)

Description:

Before confirming any new course enrollment, the system automatically validates that the student's total registered credits for the current semester will not exceed the university-mandated maximum of 15 credit hours.

Total credits ≤ 15 If student exceeds limit $\rightarrow$ show warning.

 **Credit Limit Exceeded — You cannot enroll in this course. Maximum allowed: 15 credits per semester.**

Feature 09: Course Enrollment System (Basic)

Description:

The Course Enrollment System is the primary registration interface of the platform. Students can add or drop courses.

Feature 10: Classroom Capacity Checker (Basic)

Description:

Every classroom in the university has a defined maximum seating capacity stored in the Classroom table. Before processing an enrollment, the system counts the number of students already enrolled in the course-section and compares it to the classroom's capacity. If the count equals or exceeds the capacity, the enrollment is rejected.

If classroom capacity = 40 System prevents enrollment after 40 students. Example: Class Full – Cannot Enroll

Feature 11: Student Grade Sheet System (Basic)

Description:

The Grade Sheet System allows students to view a formatted academic transcript for any completed semester.

Example Output:

Course Name	Credits	Grade	Grade Point
Intro to Programming (CSE110)	3	B-	2.70
Differential Calculus (MAT120)	3	A	4.00
English Composition (ENG101)	3	A	4.00
Semester GPA	9	—	3.57

Feature 12: Exam Schedule System (Basic)

Description:

The Exam Schedule System provides students with a view of all their upcoming midterm and final examination dates, times, and room for the current semester.

Example Output:

Course	Exam Type	Date	Time	Room
CSE110	Midterm	15 July	10:00 AM	9A-02C
MAT120	Midterm	18 July	2:00 PM	9A-05C
CSE110	Final	20 Aug	10:00 AM	9A-02C

Feature 13: Semester Fee Payment Status (Basic)

Description:

The Fee Payment Status feature allows students to view the payment record for each semester's tuition and fees.

Example Output:

Semester	Amount (BDT)	Payment Date	Status
Spring 2025	100,000	01 Jan 2026	Paid
Spring 2026	100,000	—	Pending

Feature 14: Personal Academic Routine (Weekly Class Schedule) (Basic)

Description:

The Personal Academic Routine feature generates a student-specific weekly timetable based on their current course enrollments.

Example Output:

Day	Course	Start	End	Room
Sunday	Intro to Programming (CSE110)	10:00 AM	11:30 AM	9A-02C
Monday	Differential Calculus (MAT120)	12:00 PM	1:30 PM	9A-05C
Tuesday	Intro to Programming (CSE110) – Lab	2:00 PM	3:30 PM	Lab 9B-02L
